

MAHESH BABU NARNE

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PROFESSIONAL SUMMARY

Senior Software Engineer with 4+ years building scalable distributed backend systems and consumer-facing full stack products in Java, Python, and TypeScript. Strong foundation across the full development lifecycle: API design, data modeling, event-driven pipelines, frontend interfaces, cloud infrastructure, and production operations. Has owned features end to end across communications and healthcare domains, working closely with product managers, designers, and data scientists to ship things that have measurable impact on users. Writes clean, well-tested code, takes on-call seriously as part of owning what gets shipped, and consistently looks for ways to improve systems and help the engineers around them grow.

TECHNICAL SKILLS

Languages: Java , Python, TypeScript, JavaScript, Kotlin, C#, SQL, Bash

Backend and APIs: Spring Boot, REST APIs, GraphQL, gRPC, microservices, event-driven architecture, high-availability distributed systems, API design and versioning

Frontend: React, Angular, TypeScript, HTML, CSS, component architecture, consumer-facing web application development

Messaging and Streaming: Apache Kafka, SQS, event-driven pipelines, asynchronous processing, real-time data delivery

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch, schema design, query optimization, distributed caching

Cloud and Infrastructure: Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP), Kubernetes, Docker, Terraform, CI/CD, GitHub Actions, Jenkins

AI and Automation: LLM integration, AI-assisted development (GitHub Copilot, Claude Code), agentic workflows, ML feature integration

Observability: Prometheus, Grafana, OpenTelemetry, distributed tracing, structured logging, SLO-driven alerting, on-call operations

Engineering: System design, data structures, algorithms, technical documentation, code review, cross-functional collaboration, mentorship, Agile

PROFESSIONAL EXPERIENCE

Senior Software Engineer

July 2024 – Present

Apple

United States

- Built and operated high-scale Java backend services for a consumer communications platform delivering email, SMS, and push notifications to millions of daily users, owning features end to end from API design and data modeling through deployment, monitoring, and on-call response.
- Designed REST and GraphQL APIs consumed by multiple product teams, investing in schema stability and backward compatibility so that downstream teams could ship independently without coordinating every change with the platform.
- Built React and TypeScript frontend interfaces for internal tooling and consumer-facing notification preference management, working closely with product and design to deliver experiences that matched what users actually needed.
- Developed Kafka-based event pipelines for real-time message delivery and engagement tracking, designing partition strategy and consumer group isolation to keep latency predictable for time-sensitive notifications under peak load.
- Integrated ML model outputs and LLM-powered features into production Java services, designing the integration layer to handle inference latency and partial failures gracefully without degrading the core user experience.
- Set up observability across services using Prometheus, Grafana, and OpenTelemetry structured around SLOs and real failure categories, giving the team actionable signals rather than metric noise.
- Found and fixed a performance bottleneck in the message routing layer that cut p99 latency by 35% and improved throughput by 50% after profiling showed the actual slow path rather than the assumed one.
- Mentored two engineers through complex platform features, helping them reason through failure modes and data modeling trade-offs before discovering them in production.

Software Engineer

March 2021 – November 2023

Deloitte

Hyderabad, India

- Built Java and Spring Boot backend services for SmartHealth360, a multi-tenant healthcare platform on AWS, with strict data correctness, compliance, and reliability requirements that shaped architecture decisions from the start.
- Designed REST APIs and data models for SmartHealth360's patient-facing and clinical workflows, collaborating with product owners and clinical stakeholders to translate business requirements into working software.
- Developed Kafka-based event streaming pipelines processing millions of clinical and operational events per day, applying partition strategy and consumer group design to maintain throughput and ordering guarantees under variable load.
- Optimized PostgreSQL and MongoDB schemas and Redis caching layers for SmartHealth360's multi-tenant workloads, applying indexing strategies that kept query performance predictable as data volumes and concurrent users scaled.
- Built React frontend components for SmartHealth360's clinical management workflows, working across the full stack and delivering features that worked correctly for real healthcare scenarios rather than just passing acceptance criteria.
- Diagnosed production incidents on-call by tracing failures through distributed logs and metrics to root causes, reducing recurring incidents by 40% over the course of a year.
- Led cross-team technical coordination across 10+ quarterly releases in Agile squads, consistently finding that aligning on interface contracts before implementation was harder and more important than the engineering execution itself.

KEY ACHIEVEMENTS

- Built a high-scale backend platform adopted by three product teams within two quarters, with the API and data model design flexible enough to absorb new requirements without structural changes because the interfaces were designed with future consumers in mind from the start.
- Reduced new integration onboarding time from weeks to days by standardizing how services were structured, validated, and tested, giving consuming teams a predictable contract rather than a custom process each time.
- Improved mean time to detection by 65% and mean time to resolution by 45% on a production distributed system by building observability around actual failure categories rather than raw infrastructure thresholds.
- Reduced on-call response time by 60% by writing runbooks grounded in how the system actually failed, making them useful to any engineer on rotation without requiring deep system context.
- Delivered 3 cross-functional platform projects ahead of schedule by surfacing dependency and integration risks during design rather than discovering them mid-delivery.
- Mentored 4 engineers to mid-level through consistent design feedback and code review focused on building judgment about trade-offs rather than just reviewing whether implementations were correct.

EDUCATION

Master of Science in Computer Science

University of Central Missouri, Warrensburg, MO

Bachelor of Technology in Computer Science

GITAM University, Andhra Pradesh, India

CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert

Microsoft Certified: DevOps Engineer Expert

AWS Certified Solutions Architect – Associate

Certified Kubernetes Administrator (CKA)